Customer Segmentation, Churn Analysis & CLV Modelling Using SQL and Tableau

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Tools Used:
• SQL (SQLite)
• Tableau

Introduction

This project analyzes customer behavior using RFM segmentation, churn analysis, and customer lifetime value (CLV) modelling on an Instacart dataset. The goal was to identify customer retention patterns, high-risk customer groups, and valuable customer segments using SQL and Tableau.

Dataset Overview

The dataset contains customer order activity from a public Instacart customer order dataset, including over 206,000 customers and transactional order data. SQL was used for data preparation and transformation, while Tableau was used for data visualization and dashboard development.

Source:
https://www.kaggle.com/datasets/brendanartley/simplifiedinstacartdata?resource=download

1. RFM Analysis section

Customers were segmented using RFM analysis based on:
• Recency: how recently a customer placed an order
• Frequency: how often a customer ordered
• Monetary proxy: total number of orders

*Due to the absence of transaction revenue data, total order frequency was used as a proxy for monetary value.

Customers were grouped into five segments:
Champion, Loyal Customer, Potential Loyal, At Risk, and Lost.

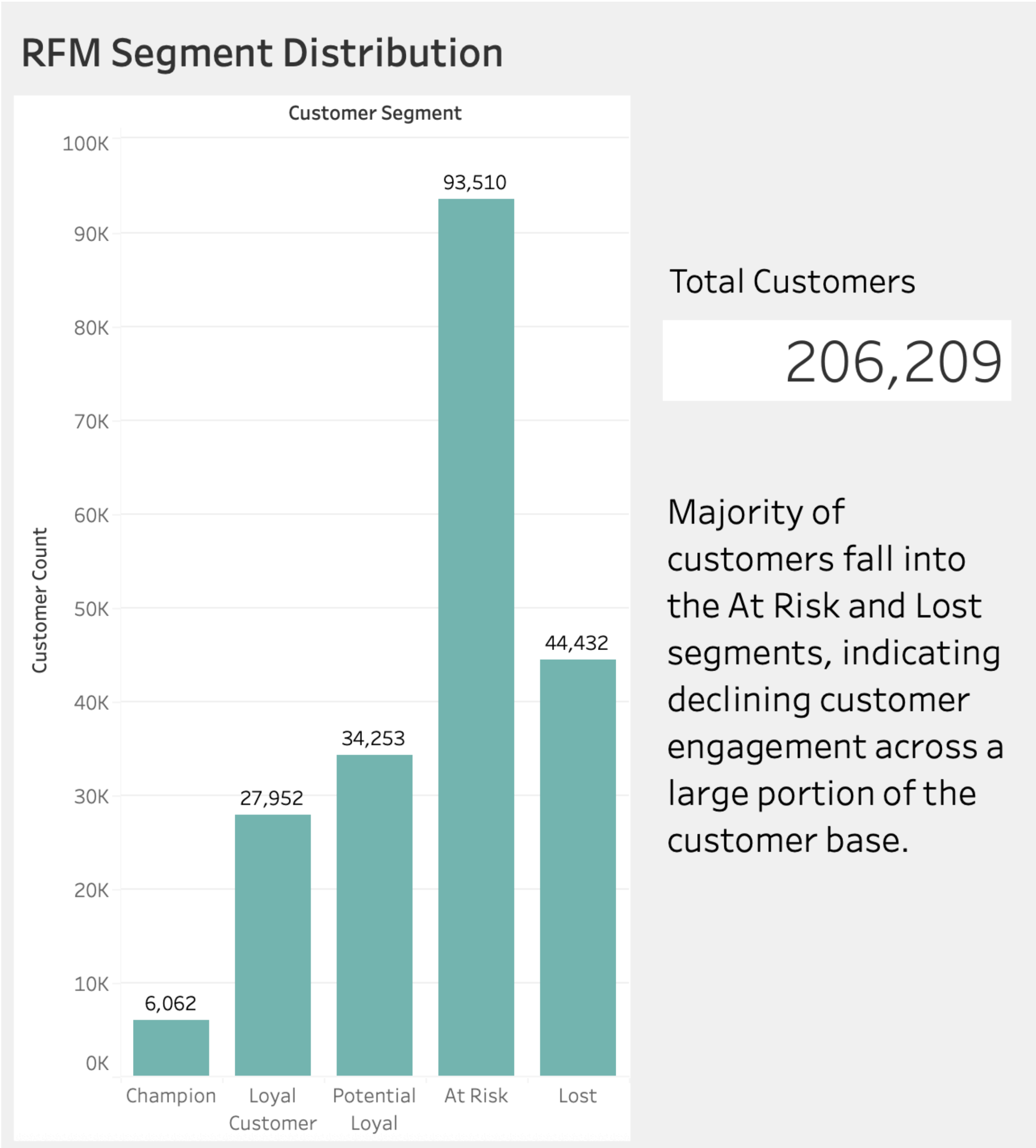
Champion - orders often and very recently
Loyal Customer - orders regularly
Potential Loyal - showing good habits but not consistent yet
At Risk - used to order but slowing down
Lost - hasn't ordered in a long time

SQL Snippets:

```
SELECT
    user_id,
    freq,
    recency,
    CASE
        WHEN recency BETWEEN 0 AND 74 THEN 5
        WHEN recency BETWEEN 75 AND 148 THEN 4
        WHEN recency BETWEEN 149 AND 222 THEN 3
        WHEN recency BETWEEN 223 AND 296 THEN 2
        ELSE 1
    END as recency_score,
    CASE
        WHEN freq BETWEEN 80 AND 100 THEN 5
        WHEN freq BETWEEN 60 AND 79 THEN 4
        WHEN freq BETWEEN 40 AND 59 THEN 3
        WHEN freq BETWEEN 20 AND 39 THEN 2
        ELSE 1
    END as frequency_score
```

```
FROM (
    SELECT
        user_id,
        COUNT(order_id) as freq,
        ROUND(JULIANDAY((SELECT MAX(timestamp) FROM orders)) -
              JULIANDAY(MAX(timestamp))) as recency
    FROM orders
    GROUP BY user_id
);
```

```
    CASE
        WHEN recency_score + frequency_score >= 9 THEN 'Champion'
        WHEN recency_score + frequency_score >= 7 THEN 'Loyal Customer'
        WHEN recency_score + frequency_score >= 5 THEN 'Potential Loyal'
        WHEN recency_score + frequency_score >= 3 THEN 'At Risk'
        ELSE 'Lost'
    END as customer_segment
```



Key Findings:

•The At Risk segment represented the largest portion of customers, suggesting a widespread decline in customer engagement and significant retention risk.
• Lost and At Risk customers represented over half of the customer base.

• Champion customers represented a small but highly engaged customer base, reinforcing the common retail pattern where a small group of loyal users drives disproportionate business value.
• The large Potential Loyal segment suggests strong opportunities for retention and loyalty initiatives to increase long-term customer value.

2. Churn Analysis section

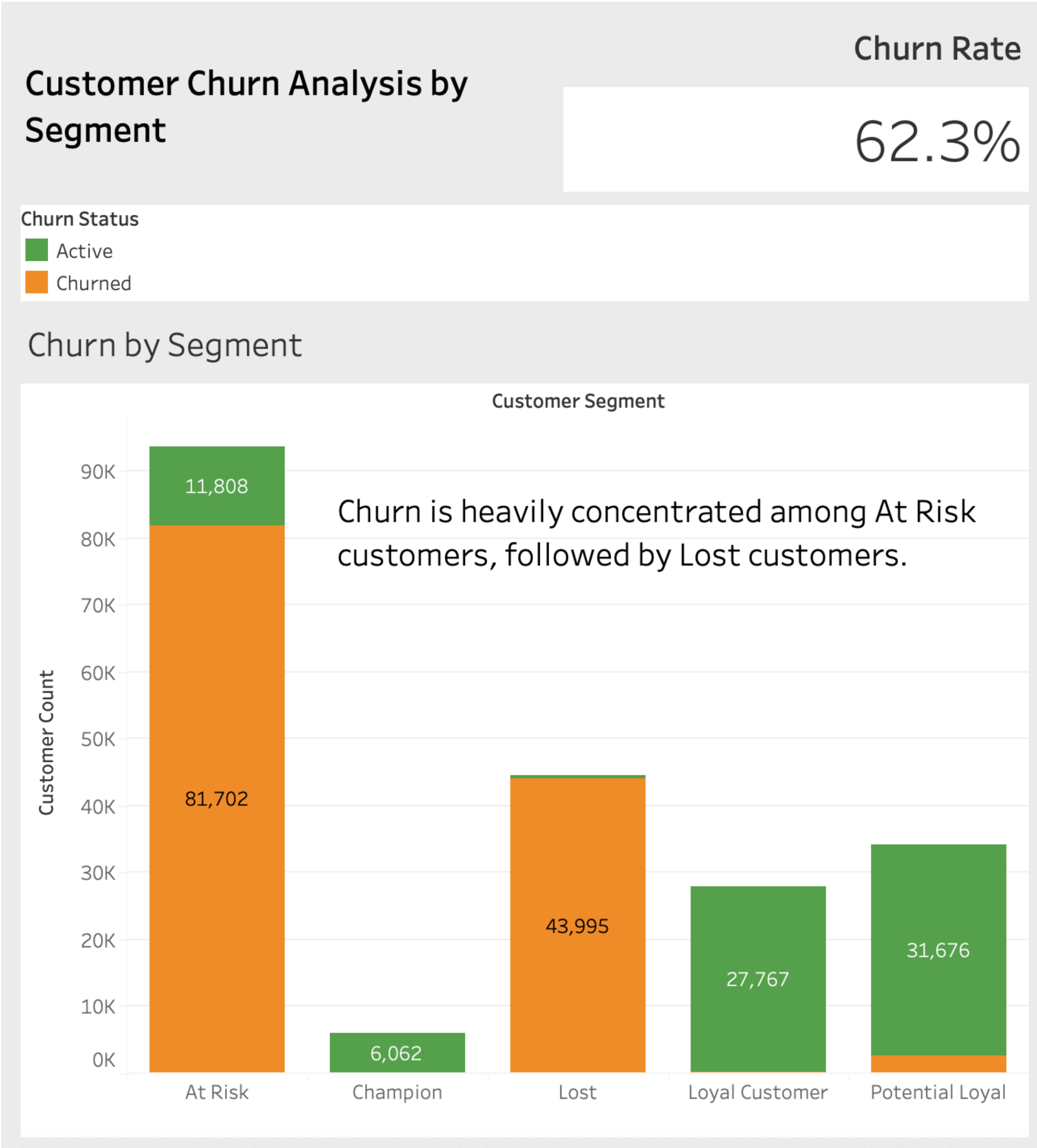
Churn analysis helps identify customers who may no longer be actively engaging with the platform. Understanding churn patterns can support customer retention strategies and reduce long-term customer loss.

In my analysis, customers were classified as churned if they had not placed an order within **180 days** of the latest transaction date.

SQL Snippet:

Created view `churn_view`:

```
CREATE VIEW churn_view AS
SELECT
    user_id,
    days_since_last_order,
    CASE
        WHEN days_since_last_order > 180 THEN 'Churned'
        ELSE 'Active'
    END AS churn_status
FROM (
    SELECT
        user_id,
        (julianday((SELECT MAX(timestamp) FROM orders)) -
julianday(MAX(timestamp))) AS days_since_last_order
    FROM orders
    GROUP BY user_id
);
```



Key Findings:

- The churn rate was **62.3%** (many users stopped engaging with the platform over time).
- Churn was heavily concentrated among At Risk customers, highlighting declining engagement as a strong indicator of customer attrition.
- Champion customers showed strong retention with no churn observed.

3. CLV Analysis section

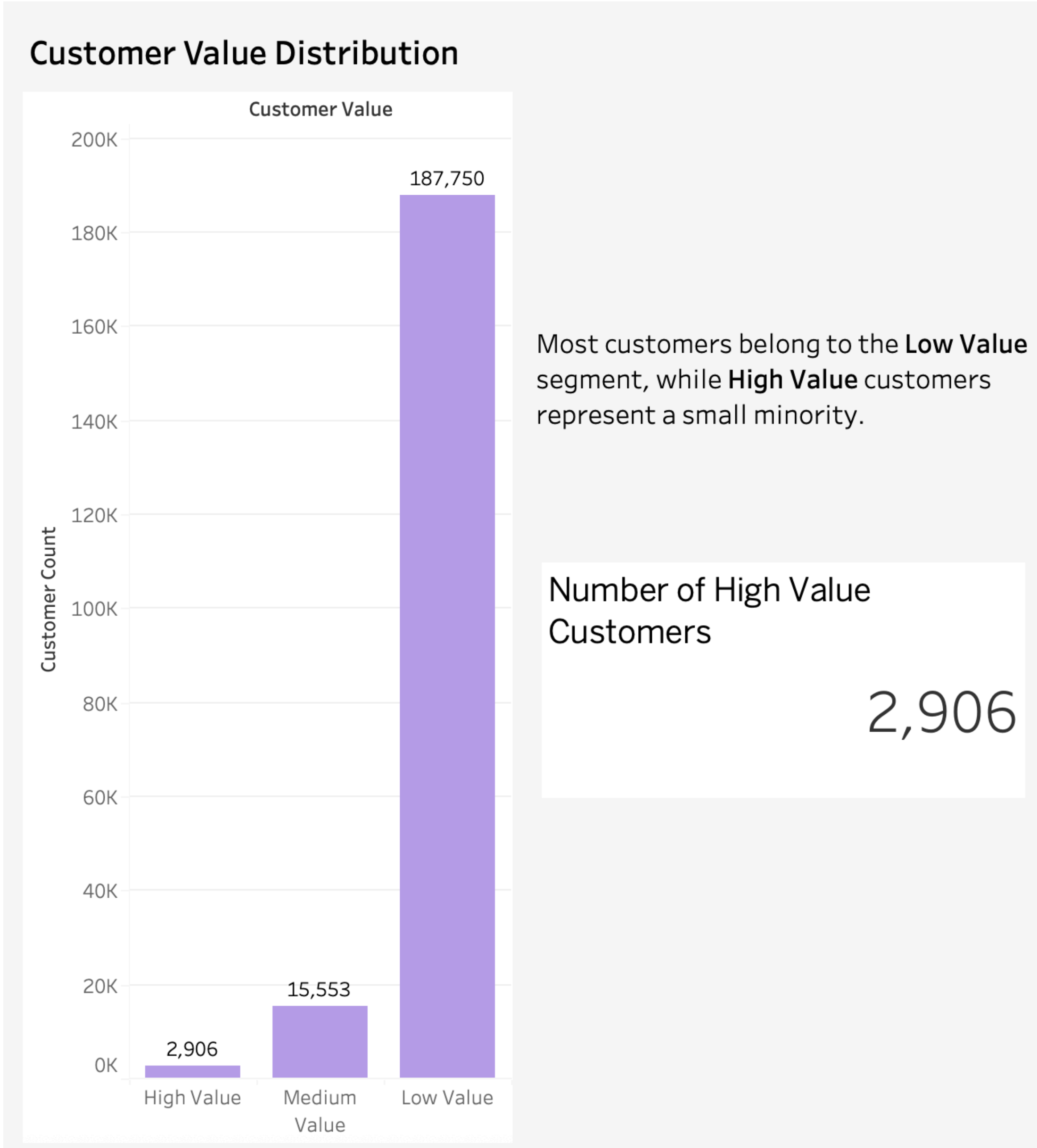
Customer lifetime value (CLV) analysis helps identify which customer groups contribute the greatest long-term value to the business. Understanding customer value distribution can support retention priorities, loyalty strategies, and marketing decisions.

Customer lifetime value tiers were estimated using total order frequency as a proxy for long-term customer value.

`SQL Snippet:`

Created view `clv_view`

```
CREATE VIEW clv_view AS
SELECT
    user_id,
    COUNT(order_id) AS total_orders,
    CASE
        WHEN COUNT(order_id) >= 80 THEN 'High Value'
        WHEN COUNT(order_id) >= 40 THEN 'Medium Value'
        ELSE 'Low Value'
    END AS customer_value
FROM orders
GROUP BY user_id;
```



Key Findings:

• The majority of customers belonged to the Low Value segment, indicating that long-term customer value was concentrated among a relatively small portion of users.

• High Value customers represented a small but strategically important segment likely responsible for a disproportionate share of repeat purchasing activity.

• Medium Value customers represent opportunities for repeat purchase growth.

Business Recommendations

- Develop retention campaigns targeting At Risk customers through *personalized discounts, reminder emails, or limited-time offers.*
- Maintain engagement with Champion and Loyal customers through *loyalty rewards, exclusive promotions, and early access incentives.*
- Encourage Medium Value customers to increase repeat purchase frequency through *targeted product recommendations and bundled offers.*
- Implement win-back campaigns for Lost customers using *re-engagement emails, special return discounts, or personalized marketing outreach.*

Conclusion

This project demonstrated how SQL and Tableau can be used to analyze customer behavior, identify churn risks, and evaluate customer value. The findings highlighted the importance of customer retention strategies and the impact of highly engaged customer segments.